PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: bukit by dan

Version: 2.0

1. Project Overview

- **Problem Statement:** Malaysians citizens across all three income categories: B40, M40 and T20, find it hard to search for the suitable government aids, subsidies, and scholarships for them. This is because the information regarding these are scattered around the internet, making it hard to determine the eligibility to apply for them. While most Malaysians would learn about national subsidies like MyKasih, some people wouldn't know the existence of for example, BANTU Sabah.]
- **Target Domain:** Enabling widespread accessibility towards information about subsidies for those who do not have the opportunity to access these resources.
- **Proposed Solution Summary:** bukit is an AI-driven website to help gather the suitable subsidies that the user needs. After inputting the details, for example, the type of subsidies, the user will just have to ask simple questions to receive tailored responses showing the exact type of subsidies they should be looking for.

2. Background & Business Objective

- **Background of the Problem:** While state governments host their own websites, the portals are usually incomplete, outdated or just hard to navigate. Furthermore, in regions with distinct geographical areas, such as Sabah, the digitalization of public services is even worse. The citizens in these areas are often forced to travel long distances to government offices just to get the paper manuals or to apply. Due to this reason, some people also do not get to know about some of the subsidies offered in their own region.

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- **Importance of Solving This Issue:** Optimizing financial relief is important in the current economic climate. A lot of money and funds are used and spent every year for welfare, but the efficiency of this is stopped by the issue. By reducing the time it takes to discover them from hours to minutes, *bukit* ensures that government funds actually reach the intended audience.

- **Strategic Fit / Impact:** bukit aligns with Malaysia's Madani Economy framework and broader digitalization efforts. By utilizing LLMs to act as an accessible citizen support agent, this project modernizes public service delivery.

3. Product Purpose

3.1. Main Goal of the System: The primary objective of bukit is to serve as an accessible AI powered conversational website that helps access to federal and state level subsidies in Malaysia.

3.2. Intended Users (Target Audience):

- Malaysian Citizens
- Families looking for financial relief
- Students
- elderly

4. System Functionalities

4.1. Description: bukit's system operates as an interactive AI for government financial aid. Instead of forcing Malaysian citizens to manually search through complex or incomplete government portals, *bukit* simply record the users' inputs and data, process and instantly render a personalized result containing the specific subsidies they are suitable for.

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4.2. Key Functionalities:

- **Smart Subsidies Matching:** The system filters out the irrelevant data and groups for the user.
- **User Friendly:** The AI transforms the complex data and information into words simpler for users to understand and read, shifting from heavy text into a summarized version.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: The system utilizes ILMU-GLM-5.1. This model was chosen for its strong natural language processing ability and its ability to output highly structured data, along with its advantage of understanding Malay documents.

4.3.2. **Prompting Strategy:** Following Google's strategy of crafting a good prompt for the system instructions, by following the format of Persona (the role the AI system will play), Task (step by step instructions), Context (information or details about the purpose of the task), Format (details on the type of the output - JSON). We have also included further context like the category of request, context string from RAG, and user details such as their birth year, income category, state of residence.

4.3.3. **Context & Input Handling**: We have set default return of context strings from RAG to be 3, and limited the total response tokens maximum to be 10k per request. This is more than sufficient as most responses should fit within the 5k window. Requests exceeding the 10k will automatically be responded to on the user interface prompting an error.

4.3.4. **Fallback & Failure Behavior:** A strict RAG system supplies the context, and the prompt limits it to only return subsidy recommendations based on the context. We have tested multiple times, and for topics that are even slightly out of range from asking for available subsidies, it will fall back to saying it can only answer about subsidies.

5. User Stories & Use Cases

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- **User Stories:** *Example:* "As an M40 head of household, I want to ask about grocery and utility assistance so that I can quickly identify targeted federal aids (like MyKasih) to help manage the rising cost of living."
- **Use Cases (Main Interactions):**

  Natural Language Query & Smart Matching: The user types a simple question into the prompt bar (e.g., "What help can I buy for groceries?"). The system merges this with their onboarding data, queries the LLM, and retrieves matches based on the criteria.

  UI Rendering: Instead of returning a long wall of text, the frontend analyses the response and instantly renders it into boxes. The user can interact with the boxes to read descriptions and find application criteria.

- **Use Case Diagram:** *(Optional but encouraged for participants to visualize the interaction)*.

6. Features Included (Scope Definition)

- **Secure Authentication & Contextual Onboarding:** A user authentication flow with encrypted JWT, also featuring a sequence where the system captures and stores the crucial data like birth year, state and income category to personalize the users' future AI interactions.
- **Smart Subsidy Matching Engine:** ILMU-GLM-5.1 utilizes context injected prompting to match users with specific subsidies (e.g., MyKasih, BUDI Sabah) based on their stored onboarding profile.
- **News Module**: A dedicated page that pulls in current subsidy updates for Malaysian citizens.

- **Categorized Subsidy Dashboard**: A discovery page that organizes financial aids into 6 sections: Utilities, Transportation, Healthcare, Education, and Groceries. This allows the users to browse general aid information visually and easily before moving into a personalized chat.

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- **High-Impact Landing Page**: A professional entry point that could gain trust by highlighting the platform's core values: real time subsidy tracking, enforced data privacy and a deep focus on the Malaysian economic context (e.g., MyKasih and RON 95 petrol subsidies).

7. Features Not Included (Scope Control)

- **Storage of Highly Sensitive Personal Data**: The system does not store IC numbers or official financial records in the database. Handling such data is too legally complex and carries significant security risks for an early stage prototype.
- **Full Profile Management**: Users cannot currently edit their profiles once created. This keeps the data flow to the AI consistent.
- **Persistent News API Caching**: While the dashboard shows all the categories, the backend does not yet support a local storage cache for a live News API. This feature is reserved for a production environment to optimize performance and token costs.
- **Local Storage Caching for News API**: While the news page is functional, the system does not yet cache this data in local storage for offline or high-speed persistent access.

8. Assumptions & Constraints

- **LLM Cost Constraint:** We estimate an average token cost of ~4,000 to 5,000 tokens per user session.
- **Technical Constraints:** The current ILMU-GLM-5.1 model is restricted to text only. This makes file attachment features (e.g., uploading PDFs of pay slips or screenshots of utility bills) unsupported.
- **Performance Constraints:** To generate our high quality results, the LLM has to output data in a complex JSON rather than standard text. There is a generation limit: the AI is constrained to return a maximum of 3 to 5 highly targeted subsidy matches per prompt.
- **User Input:** The model requires manual human prompts to retrieve data. Also, it deals with sensitive financial data, bukit is not ready for direct production level processing.

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Bukit acts strictly as an assistive discovery engine which requires human approval, meaning the users must manually verify the AI's recommendations.

9. Risks & Questions Throughout Development

- **Formatted Output:** How do we ensure the GLM/LLM is exactly returning a JSON that we can use to create our individual cards of responses?
- **Active API Connection:** How do we deal with errors in case GLM returns a 504 error?
- **High Latency:** How to prevent users from double submitting prompts in case of high latency?

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